📓 Lab Assignment: Impact of Social Media on Health

🎯 Objective

Analyze the dataset to understand the impact of social media usage on:

- Academic performance
- Mental health
- Sleep patterns

Apply:

- NumPy, Pandas
- Visualization (Matplotlib, Seaborn, Plotly)
- Statistics (T-test, Chi-square, ANOVA, p-value)
- Data Preprocessing (Scaling, Encoding)

📂 **Dataset Description**

Dataset contains:

- Student demographics
- Social media usage behavior
- Academic, sleep, and mental health indicators

🧪 **SECTION A: Data Exploration (Pandas + NumPy)**

1. Load the dataset and display:

- First 5 rows
- Dataset info
- Summary statistics

2. Identify missing values and handle them appropriately

3. Compute:

    - Mean, median, standard deviation of Avg_Daily_Usage_Hours

4. Group data by Academic_Level and calculate:

- Average usage hours
- Average sleep hours

5. Generate correlation matrix between:

    - Usage hours, Sleep hours, Mental health

📊 SECTION B: Visualization

6. Create a **heatmap** for correlation matrix
7. Plot a **boxplot**:
    - Mental health vs Overall impact
8. Create a **scatter plot with regression line**:
    - Usage vs Sleep
9. Create a **bar chart**:
    - Platform vs Academic performance
10. Build a **Plotly interactive visualization**:

- Usage vs Mental health

📈 SECTION C: Statistical Analysis

11. **T-Test**

- Compare mental health scores:
    - High usage (>5 hrs) vs Low usage (≤5 hrs)
- Interpret p-value

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12. **Chi-Square Test**

- Test relationship between:
    - Platform used and Academic performance
- Interpret result

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13. **ANOVA Test**

- Compare mental health across:
    - School, College, University
- Interpret p-value

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14. **Probability**

- Compute:
    - P(High Usage AND Poor Mental Health)
    - P(Poor Mental Health | High Usage)

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15. **Z-score**

- Detect outliers in usage hours
- Analyze their impact

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16. **Confidence Interval**

- Calculate 95% CI for sleep hours

⚙️ **SECTION D: Data Preprocessing**

17. Apply **Standardization (Z-score scaling)**:

- On usage and mental health

18. Apply **Min-Max Normalization**:

- On sleep hours

19. Compare:

- Standardization vs Normalization (with plots)

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20. Apply **Label Encoding**:

- Gender
- Overall Impact

21. Apply **One-Hot Encoding**:

- Most Used Platform

22. Compare dataset shape before and after encoding

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23. Build a **combined preprocessing pipeline**:

- Scale numerical features
- Encode categorical features

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24. Detect and handle outliers after scaling

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25. Apply **log transformation** (if needed) and compare distribution

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🧠 **SECTION E: Conceptual Questions**

26. Explain:

- Correlation vs Causation

27. When to use:

- Standardization vs Normalization

28. Why One-hot encoding is preferred over label encoding?

29. Why scaling is important for ML models?

🚀 **SECTION F: Advanced (Optional)**

30. Build a simple model to predict:

- Mental Health Score

31. Perform clustering:

- Segment students into usage groups

32. Recommend optimal usage time based on analysis

📌 **Submission Requirements**

- Code file (.ipynb / .py)
- Output plots
- Statistical interpretation
- Final insights